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B.Tech. Degree IV Semester Examination April 2018

ME 15-1404 APPLIED THERMODYNAMICS (2015 Scheme)

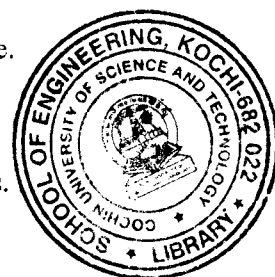
Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Briefly explain two statements of second law of thermodynamics.
- (b) State and prove Carnot's theorem.
- (c) Establish the inequality of Clausius.
- (d) Explain Rankine cycle with the help of a T-S diagram.
- (e) Explain reheat cycle.
- (f) Write a note on the effect of friction on critical pressure ratio in a steam nozzle.
- (g) Explain Wilson line in connection with super saturated flow in nozzles.
- (h) Enumerate different losses in steam turbine.
- (i) Derive an expression for the mean value of gas constant for a mixture of gases.
- (j) Write a note on adiabatic mixing of streams.



PART B

(4 × 10 = 40)

- II. (a) Derive steady flow energy equation and apply the SFEE to flow through a nozzle, diffuser, turbine, compressor and throttling device. (5)
 - (b) A turbine, operating under steady-flow conditions, receives 4500 kg of steam per hour. The steam enters the turbine at a velocity of 2800 m/min, at an elevation of 5.5 m and a specific enthalpy of 2800 kJ/kg. It leaves the turbine at a velocity of 5600 m/min, at an elevation of 1.5 m and a specific enthalpy of 2300 kJ/kg. Heat losses from the turbine to the surroundings amount to 16000 kJ/hr. Determine the power output of the turbine. (5)
- OR**
- III. (a) Derive Maxwell equations and Tds equations from fundamentals. (5)
 - (b) An insulated cylinder of volume capacity 4 m³ contains 20 kg of nitrogen. Paddle work is done on the gas by stirring it till the pressure in the vessel gets increased from 4 bar to 8 bar. Determine change in internal energy, work done, heat transferred and change in entropy. (5)
- IV. (a) Explain five methods to improve efficiency of Rankine cycle. (5)
 - (b) A steam power plant operates in a Rankine cycle with superheated steam. The inlet steam conditions are 100 bar, 550°C. The condenser operating pressure is 0.05 bar. Determine the efficiency of the cycle. (5)
- OR**
- V. (a) Draw T-S diagram of water and briefly explain steam generation process. (4)
 - (b) A simple Rankine cycle works between pressure 28 bar and 0.06 bar, the initial condition of steam being dry saturated. Calculate the cycle efficiency, work ratio and specific steam consumption. (6)

(P.T.O.)

- VI. (a) Dry saturated steam at a pressure of 12 bar flows through the nozzles at the rate of 4.5 kg/s and discharges at a pressure of 1 bar. The loss due to friction occurs only in the diverging portion of the nozzle and its magnitude is 10% of the total isentropic enthalpy drop. Determine the cross sectional area at the throat and exit of the nozzle. (6)

- (b) Derive an expression for the mass of steam leaving a nozzle. (4)

OR

- VII. (a) In a De-lavel turbine, the steam enters the wheel through a nozzle with a velocity of 500 m/s and at an angle of 20° to the direction of motion of the blade. The blade speed is 200 m/s and the exit angle of the moving blade is 25° . Find the inlet angle of the moving blade, exit velocity of the steam and work done per kg of steam. (6)

- (b) Write a short note on velocity compounding of an impulse turbine. (4)

- VIII. (a) Derive an expression for the equivalent molecular mass of a mixture of gases. (4)

- (b) A mixture of 1 mole of methane and 4 moles of air contained in a vessel is at 1 bar and 15°C . The volumetric analysis of the air can be taken as 79% N_2 and 21% O_2 . Calculate for the mixture (6)

- (i) mass of methane, oxygen and total mass.
- (ii) apparent molecular mass.
- (iii) specific volume of the mixture.

OR

- IX. (a) 2 kg of oxygen at 1.5 bar and 40°C is mixed with 4 kg of nitrogen at 1 bar and 140°C to form a mixture. The process occurs adiabatically in a steady flow apparatus. Calculate the final temperature of the mixture. For nitrogen and oxygen, specific heat C_p may be taken as 1.04 kJ/kg K and 0.95 kJ/kg K respectively. (6)

- (b) Explain Dalton's law of partial pressure, Amagat's law of partial volumes and Gibbs-Dalton law for enthalpy, entropy and internal energy. (4)
